

◆ 16.7 Decision Selection and Execution Mechanism

16.7.1 Feasible Action Set

$$\mathcal{U}_{\text{feasible}} \subset \mathcal{U}, \quad |\mathcal{U}_{\text{feasible}}| < \infty$$

where each $u \in \mathcal{U}_{\text{feasible}}$ is physically realizable and compatible with the control operator $B(u)$.

Thus:

decisions are implementable in practice.

16.7.2 Optimal Decision Rule

$$u_k = \arg\max_{u \in \mathcal{U}_{\text{feasible}}} D_u(k)$$

Thus:

an optimal action always exists due to finiteness of the feasible set.

16.7.3 Tie-Breaking Rule

$$u^* = \arg\min_{u \in \text{ties}} C(u)$$

where $C(u)$ is a secondary cost function.

Thus:

ambiguity is eliminated in decision selection.

16.7.4 Execution Threshold

$$\text{execute if } D_u(k) > D_{\text{crit}}$$

Thus:

weak or uncertain decisions are filtered out.

16.7.5 Hysteresis (Anti-Oscillation Mechanism)

$$\text{switch if }$$

$$D_{\{\text{new}\}} > D_{\{\text{old}\}} + \delta + \eta \\ \sigma$$

Thus:

rapid switching and oscillations are prevented.

16.7.6 Real-Time Constraint

$$T_{\{\text{decision}\}} \leq T_{\{\text{max}\}}$$

Thus:

the decision process satisfies real-time requirements.

16.7.7 Decision Confidence Metric

$$\{\text{Confidence}\} = \\ D_{\{\text{max}\}} - D_{\{\text{second}\}}$$

Thus:

the quality and reliability of decisions can be

quantified.

16.7.8 Energy Consistency Constraint

$$\langle \nabla \mathcal{E}(\Psi_k), B(u_k) \rangle \leq 0$$

Thus:

decisions do not increase system energy.

16.7.9 Closed-Loop Injection

$$\Psi_{k+1} = \Psi_k + dt \cdot F(\Psi_k, u_k)$$

Thus:

decisions directly influence system evolution.

16.7.10 Decision Convergence

$$\limsup_{k \rightarrow \infty} D(k) = D^*$$

Thus:

decision scores stabilize over time.

16.7.11 Deterministic Limit

$$\sigma \rightarrow 0 \Rightarrow u_k \rightarrow u^*$$

Thus:

decisions converge to a deterministic optimal policy.

16.7.12 Integration with System Components

- From Section 16.6: decision scores D_u
- From Section 16.2: control enters dynamics via F
- From Section 16.3: energy constraint ensures stability

Thus:

the decision system is fully integrated within the closed-loop framework.

Closure Statement (Part VII)

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\boxed{
\text{The system produces stable, optimal, and
executable decisions.}
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SEKHEM-TSDD Theory, Mohamed Abd Elnaby,
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